

Sep. 21 - Lecture

#54

$M = \{1, 2, \dots, n, n \cdot a\}$, determine number of n -combinations
 $\# n\text{-combinations} = \sum_{k=0}^n \# n\text{-comb. of } \{1, 2, \dots, n, n \cdot a\}$
 with exactly k copies of a

$$= \sum_{k=0}^n \# \text{ of } (n-k)\text{-combinations of } \{1, 2, \dots, n\}$$

$$= \sum_{k=0}^n \binom{n}{n-k} = \sum_{k=0}^n \binom{n}{k} = 2^n \quad \square$$

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permutations of set w/ m copies of a and at most n copies of b

Ans: Let $M_k := \{m \cdot a, k \cdot b\}$, $k = 0, 1, \dots, n$

$$\# \text{ permutations of } M = \sum_{k=0}^n \# \text{ permutations of } M_k$$

Pascal's identity

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

$$= \sum_{k=0}^n \binom{m+k}{k} = \binom{m}{0} + \binom{m+1}{1} + \binom{m+2}{2} + \binom{m+3}{3} + \dots + \binom{m+n}{n}$$

$$\begin{aligned} & \binom{m+1}{0} \rightarrow \binom{m+2}{1} \rightarrow \binom{m+3}{2} \rightarrow \dots \rightarrow \binom{m+n+1}{n} \\ & \binom{m+n+1}{n} = \binom{m+n+1}{m+1} \end{aligned}$$

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#permutations with at most m a's and n b's. $\sum_{k=0}^n \binom{m+k}{k}$

$$\sum_{j=0}^m \sum_{k=0}^n \{j \cdot a, k \cdot b\} = \sum_{k=0}^n \{m \cdot a, k \cdot b\} = \sum_{k=0}^n \binom{m+k}{m} = \sum_{k=0}^n \binom{m+k}{k}$$

$$= \sum_{j=0}^m \binom{j+n+1}{j+1} \rightarrow j \rightarrow j+1 \rightarrow \sum_{j=0}^m \binom{j+n+1}{j+1} = \sum_{j=0}^{m-1} \binom{n+(j+2)}{j+2} + \binom{n+m+1}{m+1}$$

$$= \binom{n+1}{1} + \binom{n+2}{2} + \dots + \binom{n+m+1}{m+1}$$

$$= \binom{n}{0} + \binom{n+2}{2} + \dots + \binom{n+m+1}{m+1} - 1$$

$$= \sum_{j=0}^{m+1} \binom{n+j}{j} - 1 = \binom{n+m+2}{n+1} - 1 = \binom{n+m+2}{m+1} - 1$$

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(a) ways to select 6 sticks from 20 (indistinguishable)

 $\rightarrow \binom{20}{6} \rightarrow$ (should have set up stars and bars for later problems)~~ways to select sticks such that no chosen sticks are consecutive.~~

use stars and the bars principle:

 $x_1 := \# \text{ sticks to left of } 1^{\text{st}} \text{ chosen stick}$ $x_i := \# \text{ sticks between } (i-1)^{\text{th}} \text{ and } i^{\text{th}} \text{ chosen stick}$
 $\hookrightarrow i=2, 3, 4, 5, 6$

$$\text{Then } \begin{cases} x_1 + x_2 + \dots + x_7 = 14 \\ x_1, x_2, \dots, x_7 \geq 0 \end{cases}$$

By the stars and bars principle

#ways = #integral solutions =

$$\binom{20}{6} = \binom{14+7-1}{7-1}$$

(b) ways to choose sticks such that none are consecutive,

$$\text{Then } \begin{cases} x_1 + \dots + x_7 = 14 \\ (*) \end{cases}$$

$$(*) \begin{cases} x_1 \geq 0, x_2, x_3, \dots, x_6 \geq 1, x_7 \geq 0 \end{cases}$$

ways = # integral solutions to (*)

$$= \binom{(14-5)+7-1}{7-1} = \binom{15}{6}$$

(#20) Theorem $R(3,4) \leq 10$
 $\parallel$
 $R(4,3)$

Recall: $\bullet$ represents a person (Scratch work)
 $\bullet - \bullet$ represents two people that don't know each other
 $\bullet \text{---} \bullet$ represents two people that don't know each other

Proof

Let $x_1, \dots, x_{10}$ be 10 points in the plane.

Fix x_1 .

$\bullet x_1$ There are 9 lines between x_1 and $x_2, \dots, x_{10}$
 we color the 9 lines red and blue.

$\overset{0}{x_2} \quad \overset{0}{x_3} \quad \dots$

$\bullet x_{10}$ By the strong PP, either 4 of them
 are colored red
 or 6 of them are colored blue

case 1 4 of them are colored red

case 2 6 of them are colored blue